
UniRL: Trajectory-Centric Post-Training Across Heterogeneous Generative Models

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Abstract

Reinforcement-learning post-training systems share an outer loop—generate, score, assign credit, and update—yet their implementations fragment because autoregressive tokens, diffusion trajectories, multimodal values, and multi-stage rollouts carry different state and impose different execution constraints. We present UNIRL, a trajectory-centric system that places a lineage-preserving, modality-aware intermediate representation at the boundary between post-training semantics and distributed execution. A trajectory is an ordered tree of typed parts carrying user-visible values, model-facing conditions, optimization-facing segments, rewards, and the policy version that produced each generated frontier. Rollout engines implement one transformation, **Sample**→**Sample**, while logical roles, placement, tensor transport, weight publication, and phase residency determine how the transformation executes. These contracts factor model-specific replay from lineage, movement, placement, and scheduling decisions; they do not assert that the underlying objectives or every engine-topology combination are interchangeable. This manuscript derives the design from UniRL commit f5d7104 and specifies a claim-aligned evaluation covering training reproducibility, end-to-end throughput, phase cost, scaling, memory, transport, and policy lag. A source-importing CPU harness already exercises ten core IR invariants and records preliminary container-operation costs; the multi-seed training and aligned GPU systems results remain open. **TODO[data]:** Add artifact-backed headline findings only after the primary E1–E5 evidence in Section 6 is complete; mention E6 only if its controlled sweep supports a bounded-staleness conclusion.

1 Introduction

Online post-training is a feedback program. It expands inputs into policy samples, evaluates those samples, turns feedback into a learning signal, and applies one or more optimizer updates. At this level, language-model GRPO, diffusion-model policy gradients, prompt-rewrite pipelines, and tool-using agents look deceptively similar. Their data paths do not. An autoregressive update may need packed token IDs, masks, rollout log probabilities, and per-token values; a diffusion update may need a sparse set of latent states, a noise schedule, and transition statistics; a composed workflow must retain which image descended from which rewritten prompt; an asynchronous run must additionally retain which policy version generated every sample.

Existing systems have made major progress on scalable LLM post-training and flexible device mapping (Sheng et al., 2024; Hu et al., 2024; Wang et al., 2025; NVIDIA Corporation, 2025). Fully or partially asynchronous systems address rollout tail latency and accelerator idle time (Fu et al., 2025; Zhang et al., 2026; Zhu et al., 2025). Meanwhile, diffusion RL methods and multimodal frameworks broaden the optimization and model families that can be trained (Black et al., 2023; Liu et al., 2025; Xue et al., 2025; Zheng et al., 2025; Huang et al., 2026). These advances also raise the bar for a new system paper: supporting more modalities or adding an asynchronous loop is not, by itself, a sufficient technical thesis.

UNIRL starts from a narrower question that distinguishes its thesis from a feature catalogue:

Can one post-training program cross generative model families and execution topologies while retaining the structure needed for correct grouping, replay, credit assignment, transport, and policy-version control?

Its answer is to make a trajectory intermediate representation (IR) the stable semantic object, then expose execution choices as explicit contracts around that object. The IR makes a complete prompt-rooted lineage, rather than an individual row or an LLM-specific tensor dictionary, the unit of scheduling and sharding. Typed leaves distinguish decoded multimodal values from model conditions and optimization segments. Distributed methods declare how each structured field scatters, concatenates, packs, or remains shared. Dense tensors can be replaced by row-addressable references without changing the trajectory structure. This representation is passed through a small set of logical roles and can be mapped to colocated or disjoint GPU slabs.

The paper makes three source-backed design contributions and assigns each a distinct empirical burden:

- A *lineage-preserving trajectory IR* for branching AR, diffusion, composed, and environment-interacting rollouts (Section 3). E0 and E4 must show that real traces preserve ancestry, grouping, replay state, and credit; E1 and E2 then show that the same boundary supports credible learning in two model families.
- A *factorization of semantic roles from physical execution*: rollout remains **Sample** $\rightarrow$ **Sample**, while placement, field-aware dispatch, tensor transport, weight publication, and residency vary only where their contracts permit (Section 4). E3 measures full-stack efficiency against an aligned external system, whereas E5 measures UniRL’s own representation and topology costs.
- A *version-aware bounded-staleness discipline*. It fills rollout lanes dynamically, but publishes weights only at a quiescent, batch-aligned boundary and filters samples by explicit policy lag (Section 4.5). E6 separates the cost of disaggregation from the incremental effect of overlap, then measures the lag-quality-time trade-off.

This creates a claim ladder rather than one omnibus benchmark: trace invariants establish semantic correctness; AR and diffusion runs establish training validity; cross-stage grouping tests the value of retained lineage; and matched systems studies quantify the cost and operating region of the execution choices.

The current manuscript intentionally contains no end-to-end training or systems-performance claim: the audited repository contains runnable instrumentation and comparison scripts but not the raw logs needed to verify prose-reported numbers. We report only a narrowly scoped, artifact-backed CPU regression of IR operations in RQ3. This evidence boundary prevents an implementation survey from masquerading as an evaluated systems result.

2 Problem: Preserve Semantics While Changing Execution

Let an online post-training iteration consume a batch of roots X , sample trajectories with behavior policy π_v , obtain reward r , construct credit A , and update parameters θ :

$$X \xrightarrow{\pi_v} \mathcal{T} \xrightarrow{R} (\mathcal{T}, r) \xrightarrow{C} (\mathcal{T}, A) \xrightarrow{U_\theta} \theta'. \quad (1)$$

The compact notation hides three forms of heterogeneity.

Trajectory heterogeneity. The sufficient statistics for an update depend on the generative process. For AR policies, a segment may be a variable-length packed token sequence with masks, old log probabilities, values, and returns. For stochastic diffusion or flow policies, a segment may contain selected latent snapshots, a shared noise schedule, the indices of sampled transitions, and rollout or replay transition statistics. Multi-stage and agentic paths add nonuniform lineage and observations.

Role heterogeneity. Generation, reward evaluation, model replay, gradient computation, and policy publication have different dependencies and may use distinct runtime systems. Serving engines such as

vLLM (Kwon et al., 2023) and SGLang (Zheng et al., 2024) are optimized for generation, while FSDP (Zhao et al., 2023) or another training backend owns parameters and optimizer state. Moving a large policy or trajectory between them is a first-order cost.

Topology heterogeneity. A small run may time-share one GPU set between training and rollout; a larger run may separate them; a variable-duration rollout may benefit from asynchronous admission. Each choice changes memory pressure, transfer paths, synchronization frequency, and allowable policy lag.

These observations produce four requirements.

Table 1: Design requirements and the corresponding UniRL mechanism. The right column states what must be evaluated; it is not a result.

Requirement	Mechanism	Evidence required
Preserve structure	Parent-addressed parts; root-complete split, select, concat, and reward propagation	Trace invariants across AR, diffusion, and cross-stage runs
Preserve model state	Typed primitives, conditions, and AR/latent replay segments	Rollout-replay consistency and held-out integration
Change execution safely	Logical roles, placement scopes, field-aware dispatch, and tensor references	Matched-work topology and transport ablations
Control policy age	Version-stamped generated parts, lag filters, quiescent publication, and hard boundaries	Lag distributions, throughput, and time-to-quality

Non-goals. UNIRL does not erase model-specific code: a model bundle still exposes architecture-specific stages, and an algorithm still defines its replay and loss. It does not automatically discover an optimal placement. The present runtime uses a driver to sequence roles and Ray actors (Moritz et al., 2018) to host them; it is not a new cluster scheduler. Finally, source-code breadth is not evidence that all combinations are production-ready.

3 A Trajectory-Centric Programming Model

3.1 A trajectory is an ordered tree of parts

The central type is

$$\mathcal{T} = [P_0, P_1, \dots, P_L], \quad P_i = (I_i, X_i, C_i, Z_i, R_i, A_i, V_i, M_i), \quad (2)$$

where I_i are sample identifiers, X_i decoded modality values, C_i model conditions, Z_i optimization segments, R_i rewards, A_i advantages or returns, V_i a behavior-policy version, and M_i auxiliary metadata. In code, $\mathcal{T}$ is a **Sample** and each P_i is a **Part**.

Identifiers encode ancestry. Root IDs contain no lineage separator; a branch appends a child ordinal. For every non-root row j in P_i , its parent ID must occur in P_{i-1} :

$$\forall i > 0, \forall s \in I_i: \text{parent}(s) \in I_{i-1}. \quad (3)$$

The constructor checks Equation 3. The **fork** operator appends k generated children per frontier row; **observe** appends a branch-one input from an environment; and **propagate_rewards** reduces child reward to an unscored parent.

Batch operations act on complete root groups. **Sample.split()** returns one tree-complete trajectory per root, and slicing or selection is implemented by selecting these groups rather than independent rows. Consequently, group-relative advantages and cross-stage credit do not depend on an incidental flat-row ordering after a distributed shuffle.

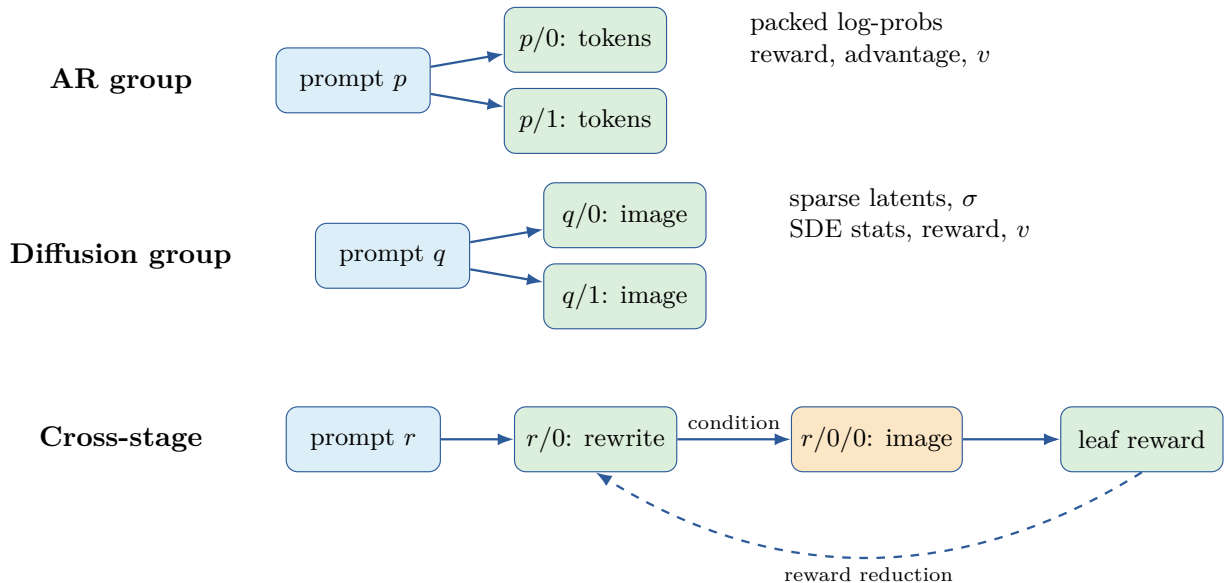


Figure 1: Three trajectory shapes represented by the same ordered-part IR. A generated part retains decoded values and the optimization segment required to replay its policy. The policy version v is attached to generated frontiers. This diagram is reconstructed from current types and execution paths, not from the repository’s legacy figures.

3.2 Values, conditions, and segments are separate layers

UNIRL distinguishes three kinds of leaves that are often conflated.

Primitives. Primitives are decoded or externally supplied values: text, images, videos, audio waveforms, and media references. Image pixels and variable-duration video/audio values use packed representations with per-example shapes or cumulative offsets. They are suitable for reward functions, visualization, and subsequent turns, but need not be the tensors replayed by the trainable model.

Conditions. Conditions are model-facing encodings, such as text embeddings or image latents. They allow rollout adapters and train-side stages to agree on typed inputs without prescribing one tokenizer, text encoder, or multimodal fusion architecture.

Segments. Segments retain optimization-facing state. **TextSegment** stores variable-length packed token fields. **LatentSegment** stores latent snapshots at selected indices, a shared sigma schedule, and optional SDE log probabilities or transition means. The sparse index map is important: a diffusion algorithm can replay only selected transitions instead of materializing an entire denoising path for every loss.

The distinction makes decoded payload lifetime independent of replay lifetime. After reward logging, trainers may drop large decoded media while retaining conditions, segments, reward, and credit for training.

3.3 A field algebra defines structural batching

Every structured batch field declares a composition rule: concatenated, packed variable-length, shared, or reduced by maximum, minimum, sum, or mean. These annotations drive concat, split, selection, repetition, and device movement recursively. They also define how a distributed method reconstructs its return value. For example, a DP-scattered rollout receives a tree-complete shard and returns parts that are merged position-wise; a broadcast lifecycle method returns one replicated control result.

This is a stronger contract than requiring each role to hand-author tensor slicing. It also creates a measurable cost: recursive structural operations and reference bookkeeping must be benchmarked in Section 6.3 rather than assumed free.

3.4 Rollout engines implement one transformation

Every rollout engine fills a request and returns a trajectory:

$$F_{\theta,e} : \mathcal{T}_{\text{request}} \longrightarrow \mathcal{T}_{\text{filled}}, \quad (4)$$

where e identifies a train-side or external generation runtime. The base contract also exposes lifecycle and policy-publication operations, but it does not standardize the engine’s internal serving API.

The current source contains four useful forms of Equation 4. The train-side adapter invokes the model pipeline directly under evaluation mode. Dedicated SGLang and vLLM-Omni adapters translate typed trajectory inputs to their native requests and rebuild typed responses. A composed engine serially fills multiple generated parts, for example an AR rewrite followed by diffusion generation. An agentic engine repeatedly appends policy turns and environment observations. The claim is therefore about a common boundary, not one universal generation kernel.

3.5 The algorithm boundary preserves specialized replay

A train stack owns microbatch planning, gradient accumulation, global loss normalization, gradient clipping, optimizer steps, and per-update aggregation. An algorithm object owns the stage-specific map

$$(C, Z, A; \theta) \longmapsto \mathcal{L} \quad \text{and backward.} \quad (5)$$

Before repeated optimizer updates, the stack can freeze an old-policy anchor at the same microbatch geometry used for training. AR objectives replay token log probabilities; diffusion objectives replay selected transitions and may compare rollout statistics, a pre-update replay, or a reference model. Supervised and forward-process objectives explicitly opt out of advantages.

This division is deliberately not advertised as algorithmic novelty. GRPO follows the group-relative formulation introduced for language reasoning (Shao et al., 2024); PPO follows proximal policy optimization (Schulman et al., 2017); diffusion objectives build on prior diffusion-RL methods (Black et al., 2023; Liu et al., 2025; Xue et al., 2025; Zheng et al., 2025). The systems question is whether these objectives can consume the same trajectory and execution contracts without hiding their different sufficient statistics.

4 From a Program to a Distributed Execution

Figure 2 separates the feedback program from the mechanisms that execute it.

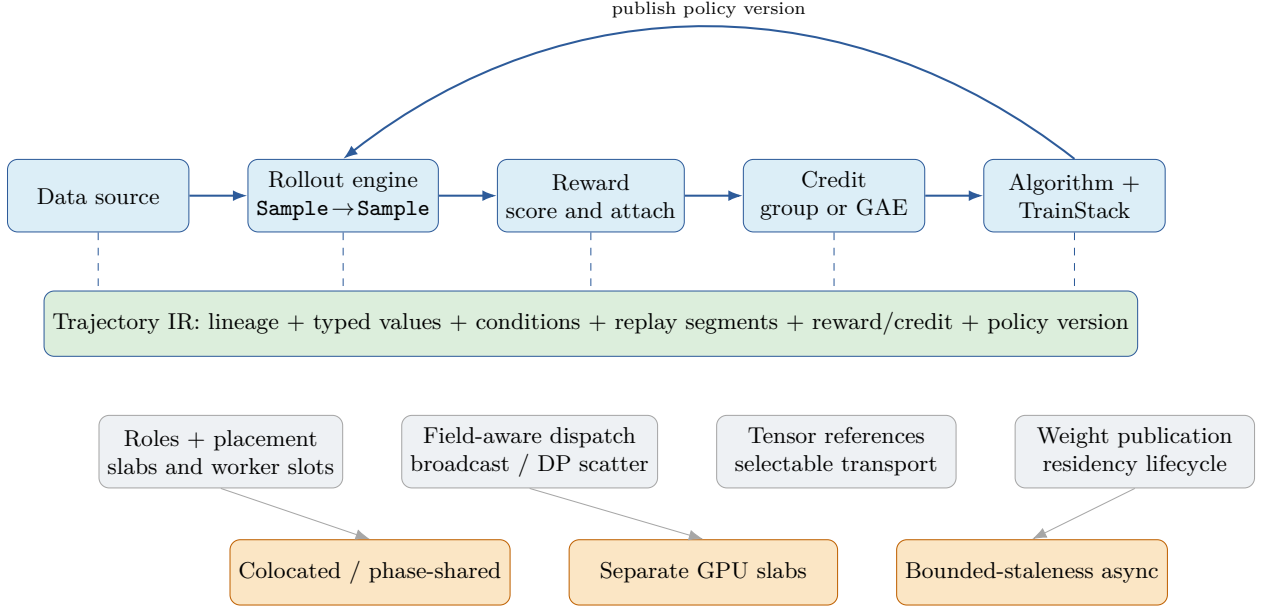


Figure 2: Source-derived UniRL architecture. The top row is the feedback program; the green bar is its stable data boundary. The lower rows map logical roles to communication and resource policies. Not every topology is valid for every engine; startup validation rejects unsupported combinations.

4.1 Logical roles, device slabs, and worker slots

A `DevicePool` creates one Ray worker per GPU initially and reserves one strict-packed placement group per node. A placement scope selects a fraction of currently available devices. Roles created inside a scope either share its worker slot—enabling local sibling access and memory time-sharing—or receive isolated slots on the same devices. Disjoint top-level scopes form separate training, rollout, or reward slabs.

A role is a regular Python object derived from `Remote`. The handle constructs one role replica on each selected worker and derives data-, tensor-, pipeline-, sequence-, and expert-parallel rank metadata from the role configuration. Method decorators state argument dispatch and return collection. The trainer is consequently a single-controller program, while each method may invoke multi-controller distributed training or serving inside a role. This resembles HybridFlow’s separation between orchestration and distributed computation (Sheng et al., 2024); UniRL’s paper claim centers instead on carrying the same typed lineage through heterogeneous generative stages.

4.2 Tensor references decouple structure from movement

Before an RPC, dense tensor leaves may be *dehydrated* into `TensorRefs`. A reference is an ordered list of half-open row spans over opaque handles, plus shape, dtype, and device metadata. Selection and contiguous slicing construct new span views without moving tensor data. At a destination, hydration resolves all referenced spans in a batch and reconstructs the original nested object.

The transport interface has a deliberately small contract: store tensors, resolve handles, identify references, and optionally localize references for destination workers. The audited source provides (i) a colocated per-worker store, (ii) a per-GPU store with IPC and NCCL transfers, and (iii) a `TransferQueue` path backed by simple storage or Mooncake-compatible storage. Mooncake itself was proposed for disaggregated serving (Qin et al., 2024); here its storage interface is a transport option rather than a claimed UniRL invention.

The driver can therefore manipulate trajectory structure and metadata without materializing all GPU tensors. Whether this reduces memory and transfer cost in practice is RQ3, not an assumption.

4.3 Phase lifecycles and policy publication

Training and generation often hold different copies or formats of a policy. UniRL exposes their interaction as an ordered lifecycle:

1. wake the rollout engine sufficiently to receive weights;
2. publish parameters if the cadence requires it;
3. optionally offload colocated training state;
4. generate and stamp the frontier with the engine’s policy version;
5. sleep the engine when the selected residency policy requires it;
6. restore training state before replay and optimization.

The concrete order varies by engine capabilities. For example, staged wake can receive weights before mapping the rest of an AR engine, and a train-side engine reads live model objects without a separate publication role. Separate slabs use an explicit one-time handshake. Weight-sync implementations cover LoRA tensor paths and several full-weight paths, including tensor serialization, CUDA IPC, NCCL, and checkpoint rendezvous. The base engine can also receive per-track updates for composed models.

The source rejects combinations whose lifecycle has not been validated, such as a train-side rollout on a physically separate slab or selected EMA/offload combinations. This fail-closed policy is important because a partially awakened engine or silently stale adapter can produce plausible but invalid training data. Section 6 therefore includes replay-consistency and failure-injection checks alongside throughput.

4.4 Synchronous execution

The synchronous AR trainer performs rollout, reward, credit assignment, replay/update, optional evaluation, and checkpointing in order. The diffusion trainer can accumulate multiple rollouts into one optimizer window and can place reward on the train slab or a third slab. In both cases, topology affects resource ownership and residency but not the trajectory boundary. This makes a controlled topology experiment possible: hold model, prompts, group size, reward, number of replayed steps or tokens, and optimizer updates constant while changing only engine or placement.

4.5 Bounded-staleness execution

Asynchrony changes correctness conditions, not just control flow. UniRL’s asynchronous AR and diffusion trainers share one manager. A background dispatcher fills every engine slot up to a configured per-worker capacity. Completed trajectories are grouped by root; only complete groups enter the ready buffer. Every generated part must carry an output policy version v_o .

Let v_t be the current optimizer-update version and u the number of optimizer updates performed per consumed rollout batch. The manager exposes lag in update units,

$$\ell = v_t - v_o, \tag{6}$$

and filters completed work against a configured budget. Before publishing version v_t , it pauses admission, drains or suspends work, completes any started or batch-aligning prefix, checks that no publication splits a training batch across behavior versions, publishes weights, updates the rollout version, and then resubmits eligible pristine carry. Evaluation and checkpointing require a fully empty manager.

This discipline differs from claiming fully asynchronous optimization. Weight publication remains a barrier, and the current asynchronous diffusion path deliberately limits in-flight batch structure more tightly than the AR path. The evaluation must report the actual lag histogram, dropped work, and barrier time; a nominal “async” label is insufficient.

5 Instantiations of the Common Contracts

This section illustrates why the trajectory boundary is useful; it is not a support matrix.

5.1 Autoregressive reasoning

An input part carries prompt text and metadata. Forking by group size creates generated shells with sibling IDs. An AR engine fills packed token segments and decoded text, a reward role attaches scalar or component rewards, and the part computes group-normalized credit or token-level generalized advantage estimates. The train stack replays the same segment and applies the selected objective. Optional token-length balancing permutes complete samples before DP sharding, preserving sibling identifiers and precomputed credit.

5.2 Diffusion and flow generation

A diffusion request forks prompts and records sampling parameters, a sigma schedule, selected SDE indices, and an initial-noise recipe. The engine returns decoded media plus a latent segment whose sparse trajectory fields are sufficient for the selected update. FlowGRPO-style objectives replay recorded steps and compare new transition log probabilities against a frozen rollout or replay anchor (Liu et al., 2025). Forward-process objectives such as DiffusionNFT use clean generated samples rather than the same reverse-trajectory likelihood path (Zheng et al., 2025). The common train stack handles microbatching and optimizer state; the algorithm remains responsible for these incompatible estimators.

5.3 Cross-stage and environment-interacting trajectories

In a prompt-enhancement path, one root produces one or more AR rewrites, and every rewrite produces one or more images. The image reward is attached to the leaf and reduced to the rewrite parent before updating the relevant stage. In a tool-interacting path, the policy emits an action-like text part, the environment appends an observation part, and the process repeats until termination or suspension. Both workflows need ancestry to survive rollout scheduling and sharding; flattening them into unrelated rows would require rebuilding that correspondence out of band.

The decisive experiment is a real cross-stage trace and a controlled change in its lineage-defined grouping objective, not the number of in-tree adapters.

6 Evaluation Design and Prespecified Protocol

We organize evaluation around four research questions. Each reported run must archive the resolved configuration, commit and dirty diff, model and dataset revisions, environment manifest, raw logs, structured metrics, parser version, and evaluation outputs. Primary comparisons use the same hardware one framework at a time. The operational commands, run-directory schema, and failure-routing rules are frozen in GPU_HANDOFF_README.md; the stricter evidence-admission contract is in EXPERIMENT_RUNBOOK.md.

Table 2: Prespecified experiment-to-claim map. E0 is a gate, not a result; P0 rows are required to close the paper’s primary trajectory-centric claim.

ID	Priority	Controlled study	Paper evidence
E0	P0 gate	Environment, semantic parity, replay/publication, checkpoint, and instrumentation checks	Evaluation validity and Appendix C
E1	P0	Qwen3-4B-Base GRPO on DAPO-Math, three seeds	RQ1 AR validity and learning curves
E2	P0	SD3.5-Medium FlowGRPO, three seeds and independent image evaluation	RQ1 diffusion validity and collapse guards
E3	P0	Aligned UniRL-VeRL-Omni SD3.5 pair on one eight-GPU node	RQ2 throughput, resources, and phase cost
E4	P0	Frozen-Qwen prompt enhancement with original-prompt versus rewrite-local grouping	RQ3 lineage and enablement result
E5	P0 minimum	IR/transport costs plus a matched UniRL topology pair	RQ3 abstraction and execution cost
E6	P1	Matched synchronous/asynchronous Qwen3 sweep	RQ4 lag-quality-time trade-off

The order is E0, then E1–E4 and the minimum E5 topology/cost slice; E6 and broader E5 scaling/transport backends follow. If compute is limited, we complete primary E0–E5 before adding more models, algorithms, or scaling points: without E4 the paper would not directly test lineage, and without the minimum E5 slice it would not bound the cost of that abstraction.

Environment fields intentionally left open. The experimental design fixes semantic work before the allocation is known, but does not guess hardware facts. The GPU model, node count, GPUs per node, NVLink/PCIe/InfiniBand topology, CPU, driver and compatibility libraries, exact local model snapshot hashes, input/evaluator hashes, AR baseline commit, and released run IDs are currently “–” and will be populated directly from admitted manifests.

6.1 RQ1: Does UniRL reproduce credible training behavior?

The first question is scientific rather than purely operational. A successful run must improve over its frozen base on independent endpoints without diagnostic collapse. An implementation reference is reported only where semantic parity is attainable; because UniRL does not claim a new optimizer, the absence of an aligned AR reference limits comparative conclusions but does not turn a valid learning run into an algorithm comparison.

AR workload. E1 trains Qwen3-4B-Base with full-parameter textbook GRPO on DAPO-Math-17k. Each rollout contains 64 prompts with eight samples (512 trajectories), uses SGLang TP=1 with thinking enabled, temperature/top- p 1.0, top- k disabled, and at most 8192 new tokens. The learner uses group-standard-deviation normalization, symmetric clipping at 0.2, **seq-mean-token-mean** aggregation, four updates per rollout, a 10,240-token microplanner, AdamW learning rate 10^{-6} and weight decay 0.01, and full-weight publication every rollout. We prespecify 800 rollouts on 32 total GPUs, internal AIME evaluation every ten rollouts, full checkpoints every 200 rollouts, and seed IDs 11, 22, and 33 applied to data order and the SGLang server RNG. Frozen base and checkpoints 200/400/600/800 are evaluated on MATH-500 (500 problems, avg@4) and AIME 2024/2025 (30 problems each, avg@16) using the registry’s fixed decoding and **math_verify** grader. The UniRL run and the selected current reference must align prompts, chat template, verifier, maximum generation length, group size, effective batch, clipping, KL treatment, optimizer updates, and evaluation decoding. The public veRL Qwen3 FSDP example is only a scaffold because its stock model variant, data, group size, response budget, and KL/objective settings differ; the AR reference row remains blank until a pinned adapted launcher passes a resolved-config parity audit. We report seed-level reward and external-evaluation curves, final and best-prespecified-checkpoint results, and 95% confidence intervals.

Diffusion workload. E2 fixes SD3.5-Medium and the same aligned FlowGRPO setting used by E3: 48 prompts with 16 images (768 samples) per rollout, 384×384 training resolution, ten denoising steps, three SDE steps in the first half, $\eta = 0.8$, guidance 1, and distinct initial noise. PickScore supplies the training

reward; LoRA rank/alpha is 32/64 on eight matched attention projections; the learner uses two updates, microbatch eight, learning rate and weight decay 10^{-4} , clip 10^{-5} , and publication every rollout. We prespecify 300 rollouts on one eight-GPU node with seed IDs 11, 22, and 33. Frozen base and every 50-rollout checkpoint are evaluated with GenEval2 and PartiPrompts HPSv3/ImageReward views independent of the PickScore training reward. A preregistered diversity guard is also required so that reward improvement cannot alone establish success.

The diversity guard is within-prompt mean pairwise LPIPS (AlexNet features), computed over a fixed 16-image set per evaluation prompt with identical image seeds for the base and checkpoint. We report the prompt-level paired change and 95% bootstrap interval; a greater than 10% relative decrease from the base is preregistered as material diversity loss. This guard diagnoses collapse and is not itself a perceptual-quality metric.

Table 3: Planned single-stage training-validity matrix. Dashes are deliberately not numerical placeholders.

ID	Workload / external endpoint	Base	Reference/control	UniRL	Seeds
E1	Qwen3-4B-Base GRPO; MATH-500/AIME	–	–	–	3
E2	SD3.5-M FlowGRPO; GenEval2/preference/diversity	–	–	–	3

TODO[data]: Populate with mean, uncertainty, exact checkpoint revisions, and released run IDs.

6.2 RQ2: What is end-to-end system performance?

E3 is the primary performance comparison: UniRL versus VeRL-Omni commit 01c87ee5 for the exact E2 SD3.5 work unit on one reserved eight-GPU node (Huang et al., 2026). Both sides use 48 prompts, 16 samples, 384×384 output, ten denoising steps, three early-window SDE steps, two optimizer updates, microbatch eight, LoRA 32/64 on the same projections, learning rate/weight decay 10^{-4} , clip 10^{-5} , and PickScore. We run three process-level repetitions per configuration, 30 steps each, remove exactly the first five timing observations, and retain at least 20 observations per repetition. Systems run one at a time on the same hosts with matched cache and clock policy. We report a backend-aligned SDPA-class row and separately each system’s preregistered best valid backend. Engine/version, old-log-probability, and exact SDE-kernel differences are inherent and disclosed, not silently normalized away. Thus E3 is a work-aligned, implementation-faithful comparison of the two complete stacks; it does not isolate framework overhead from every backend/kernel difference.

An E1 AR comparison is secondary and remains blocked until the adapted veRL launcher matches model, data, prompt/chat template, verifier, generation budget, effective batch, estimator, token packing, optimizer work, publication, and evaluation. If semantic parity cannot be achieved, the row is omitted rather than replaced by a nearby recipe.

Metrics include median, mean, and p90 iteration time after a prespecified warmup; samples/s and samples/GPU-hour; per-phase time for generation, reward, credit, anchor/replay, backward/optimizer, weight publication, wake/sleep, and barriers; peak allocated and reserved GPU memory by role; CPU memory; and utilization. Strong and weak scaling use fixed global and fixed per-GPU work, respectively, after the primary single-node pair is complete. The 30-step E3 protocol does not support a time-to-quality claim; that quantity belongs to the full-budget E6 study unless an independently preregistered, full-budget baseline-training study is added.

Table 4: Systems result schema. Every cell will be derived from archived raw artifacts.

System	Configuration	s/iter	samples/s	samples/GPU-h	peak GiB
UniRL	SDPA-class aligned	—	—	—	—
VeRL-Omni	SDPA aligned	—	—	—	—
UniRL	preregistered best valid	—	—	—	—
VeRL-Omni	FA3 best valid	—	—	—	—

TODO[data]: Run on identical hardware, disclose inherent stack-version differences, and include failed/retried iterations.

6.3 RQ3: What does the trajectory abstraction cost and enable?

RQ3 first tests the semantic capability that motivates the IR, then measures its cost. E4 uses the executable frozen-Qwen3-0.6B rewrite $\rightarrow$ SD3.5 path: eight original prompts, four rewrites per prompt, and eight images per rewrite, or 256 image descendants per rollout; only diffusion trains. The primary condition computes group-relative advantages across all 32 descendants of one original prompt, while `diffusion_group_scope=rewrite` groups the eight images under each rewrite. Both conditions must serialize redacted root/part/parent IDs, shapes, reward/advantage, replay indices, group labels, and output versions. Invalid lineage remains an E0 fail-fast test rather than an intentionally corrupted quality baseline.

This control establishes that retained ancestry selects a real optimization grouping; it does not by itself prove better task quality. The current PickScore request is conditioned by the rewritten text, so a claim about preserving the user’s original intent additionally requires lineage-keyed evaluation of every image against its root prompt. We therefore report both rewrite-conditioned training reward and frozen HPSv3/ImageReward evaluations joined to each root prompt. Deterministic trainside-AR seed plumbing, real trace dumping, and the root-prompt evaluation join are E0 prerequisites for a multi-seed E4 result. If the two correct groupings tie, the experiment still establishes semantic expressiveness but not a quality benefit.

We then measure recursive IR operations and tensor transport separately from model compute. Microbenchmarks vary root count, branch factor, number and size of tensor leaves, ragged token/media lengths, and part depth. They report split/concat/select latency, reference metadata bytes, dense bytes avoided at the driver, same-GPU and cross-GPU transfer time, cross-node time when available, and correctness hashes after reassembly.

E5 compares the colocated store, per-GPU store, and TransferQueue path where the hardware supports them. At the audited commit the full GPU transport harness is not yet checked in, so implementation and review of that harness is an explicit prerequisite rather than an implied result. We also compare whole-tree sharding against a flat-row implementation that explicitly reconstructs equivalent lineage; semantic errors are caught in E0 and are never timed as valid work.

The minimum topology slice reuses E3’s colocated SD3.5 UniRL row and adds the same work with `sd3_vllmomni_lora_separate.yaml`: four GPUs train and four generate while the total eight-GPU allocation remains fixed. This pair measures two deployment regimes, not a pure transport primitive—training parallelism, residency, and local versus remote LoRA publication change together and are reported in the phase breakdown. Broader same-/cross-GPU and cross-node backend measurements remain E5 extensions. We also report which bundle/stage/adaptor code E4 requires and which trainer, scheduler, transport, and credit modules remain unchanged; changed lines are supporting evidence, not a standalone quality metric.

Preliminary executable evidence. We ran a source-importing regression harness against a clean UniRL checkout at `f5d7104`; it does not mock the trajectory classes. All ten checks passed: cross-stage lineage construction, exact tree split/concat, whole-tree select/slice, leaf-to-root reward propagation and group advantages, rejection of malformed lineage and primitive modality/batch violations, variable-length packed-text round trips, and lazy `TensorRef` selection/materialization including bounds checking. Table 5 reports a deliberately narrow timing slice from the same run. The synthetic workload has four AR children per root, one observation per AR child, two diffusion children per observation, and a four-value image payload per leaf.

Table 5: Preliminary single-process CPU trajectory-IR timings (microseconds). Python 3.12.14 and PyTorch 2.11.0; one thread; ten warm-up and 100 measured repetitions. These are regression measurements, not distributed or GPU throughput.

Roots	Leaf rows	Operation	Median	p90
8	64	<code>sample.split_concat</code>	201.3	212.0
8	64	<code>sample.select_half</code>	954.2	977.7
8	64	<code>tensorref.select_strided</code>	20.1	20.4
32	256	<code>sample.split_concat</code>	488.6	496.8
32	256	<code>sample.select_half</code>	3538.6	3615.8
32	256	<code>tensorref.select_strided</code>	74.1	76.8
128	1024	<code>sample.split_concat</code>	1653.0	1700.5
128	1024	<code>sample.select_half</code>	13893.5	14139.3
128	1024	<code>tensorref.select_strided</code>	327.9	337.5

The current `Sample.select` path forms tree-complete root shards before gathering them, so its cost grows with the full input tree rather than only the selected rows. Strided `TensorRef` selection creates many metadata spans while contiguous selection coalesces them; no dense tensor is copied until materialization. These observations motivate, but do not replace, the planned GPU transport and end-to-end ablations. The manifest, 100 per-cell raw trials, structured check outcomes, generated table rows, and SHA-256 checksums are archived under `artifacts/trajectory_ir/cpu_f5d7104_macos_arm64/`.

6.4 RQ4: When is bounded-staleness execution useful?

E6 compares E1-derived runs with the same 32-GPU allocation, 64-by-eight rollout work unit, four optimizer updates, reward, evaluator, and 800-rollout budget; disaggregated points fix half the GPUs for training and half for rollout and keep `per_worker_inflight=14`. The colocated synchronous point S0 and disaggregated zero-overlap point D0 reveal the allocation/residency cost of separating the slabs; D0 and A1 then isolate the onset of overlap within the same disaggregated trainer and topology. After a capacity pilot, the remaining primary grid fixes `max_inflight=2` and varies publication interval and allowed lag as shown in Table 6. Only after the primary grid may in-flight capacity one and four be added at one fixed publication/lag point. Natural generation variance is stratified by prompt and response length; any synthetic-delay study is labeled as a secondary stress test. Results include generation-latency distributions, engine occupancy, trainer idle fraction, publication/barrier time, completed and discarded samples, realized lag histograms, reward/eval versus optimizer update, and reward/eval versus wall time. The key plot is a Pareto frontier of time-to-quality and realized policy lag.

Table 6: Prespecified E6 primary points. Each uses seed IDs 11, 22, and 33. Configured lag is in consumed rollout batches; one batch performs four optimizer updates.

Point	Trainer / topology	In-flight	Publish interval	Max lag (batches)	Purpose
S0	sync / colocated	–	1	0	Total-resource reference
D0	async / disaggregated	1	1	0	Same-topology no-overlap control
A1	async / disaggregated	2	1	1	Isolate onset of overlap
A2	async / disaggregated	2	1	2	Isolate lag allowance
A3	async / disaggregated	2	2	2	Isolate publication cadence
A4	async / disaggregated	2	2	4	Aggressive bounded point

A controlled pair does not yet exist in the stock recipes. At the audited commit, the nominal Qwen3 GRPO sync and async example files differ in five fields that affect either the estimator or learner execution (Table 7). Consequently, measurements from those files without overrides are inadmissible as evidence about

asynchrony. The GPU study will first choose one value for every controlled field, apply explicit overrides, archive both fully resolved Hydra configurations, and produce a field-by-field parity report. Only topology, concurrency, publication interval, and allowed policy lag may then vary.

Table 7: Mandatory preflight corrections for the stock Qwen3 GRPO sync/async pair at commit **f5d7104**. These are source/config observations, not results.

Controlled field	Sync stock	Async stock	Required action
<code>normalize_adv_by_std</code>	<code>true</code>	<code>false</code>	Match the advantage estimator.
High clipping bound	<code>unset</code>	<code>0.28</code>	Match symmetric/asymmetric clipping.
Loss aggregation	sequence mean, then token mean	sequence mean, token sum normalized	Match sequence-length weighting.
Attention implementation	<code>flex_attention</code>	<code>default</code>	Match for backend-aligned rows.
Microplanner	token budget, 10,240	default count planner	Match packing and learner work.

Measurement readiness. The current structured logger supplies end-to-end step time; six coarse phases (wake, generation, sleep, weight synchronization, reward, and aggregate training); rollout diagnostics; peak allocated/reserved GPU memory; and output, train, and published versions with realized lag. Before RQ2 and RQ4, instrumentation must separate credit, replay/anchor, backward, optimizer, checkpoint, and publication-barrier time, and must emit complete per-step series for buffer occupancy and admitted, carried, suspended, rejected, discarded, retried, and completed work. GPU utilization and power must be collected by an external monitor synchronized to the run clock. Missing structured fields may not be reconstructed from isolated console messages or dashboard screenshots.

Predeclared interpretation and escalation. Expected directions are hypotheses rather than results. For E1, independent MATH-500/AIME performance should improve from the base without collapse in length, truncation, group variance, ratio, or clipping diagnostics. For E2, PickScore should improve without material loss in GenEval2, independent preference views, or the preregistered diversity guard. For E3, competitive throughput must survive complete end-to-end and failure accounting. For E4, lineage/group membership must be exact, while a quality advantage for original-prompt grouping must be judged against root prompts and remains optional evidence rather than an invariant. For E6, moderate lag should reduce idle time and may improve time-to-quality, whereas aggressive lag may increase rejection or harm quality. Configuration, logging, parity, lineage, version, checkpoint, and replay/publication faults are engineering failures to diagnose and rerun under a new ID. Changing the optimizer, reward, estimator, formal budget, model/task, evaluator, aligned-baseline definition, or central claim requires a research decision and a newly registered experiment. Stable negative results after correctness passes are retained; they are not repaired by silent hyperparameter search.

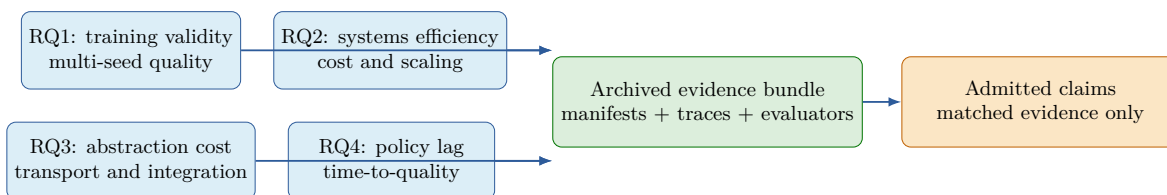


Figure 3: Claim-aligned evaluation contract. Training validity, cross-stage lineage, and system cost close the primary claim chain; bounded staleness maps the conditional operating region. Every quantitative claim must resolve to an archived evidence bundle.

TODO[data]: Add artifact-generated result panels for learning curves, phase breakdown, scaling and utilization, and the throughput-policy-lag Pareto frontier.

6.5 Fairness and statistical protocol

Effective work is matched before wall-clock time: prompts, samples per prompt, decoded resolution or maximum tokens, rollout steps, replayed steps/tokens, microbatching semantics, and optimizer updates. Reward placement is either identical and counted on both sides or excluded with a separate service-cost report. We prespecify warmup removal, use at least 20 steady-state iterations for throughput summaries, and report the full distribution rather than a single minimum. Training summaries use the seed as the statistical unit, show every seed, and report the mean with a 95% t interval; bootstrap intervals over prompts or images are labeled separately and never substituted for seed uncertainty. Systems summaries show all three process-level repetitions and the pooled per-step distribution after the fixed five-observation warm-up. Time-to-quality uses the same evaluator and a threshold fixed before comparing systems.

7 Related Work

LLM post-training systems. HybridFlow/veRL combines single- and multi-controller execution and introduces hierarchical APIs for mapping RLHF dataflows to devices (Sheng et al., 2024). OpenRLHF uses Ray, vLLM, and DeepSpeed to schedule the models in an RLHF workflow (Hu et al., 2024). ROLL emphasizes a single-controller abstraction, parallel workers, sample lifecycle scheduling, and flexible device mapping (Wang et al., 2025). NeMo-RL provides scalable training and generation backends and multimodal model support (NVIDIA Corporation, 2025). UniRL shares their concern for composable orchestration; its intended distinction is the lineage- and modality-aware trajectory boundary across AR, diffusion, and cross-stage generation. That distinction must be established by RQ3 rather than a feature table.

Asynchronous post-training. AReaL fully decouples generation and training and controls staleness to improve language-reasoning throughput (Fu et al., 2025). Relax uses service-level separation, a TransferQueue data bus, and configurable staleness for omni-modal post-training (Zhang et al., 2026). slime supports persistent asynchronous rollout generation around a training/rollout/data-buffer loop (Zhu et al., 2025). These systems make asynchrony an execution design space, not a UniRL novelty claim. UniRL contributes a particular version-stamped group and publication-boundary discipline shared by AR and diffusion paths; RQ4 tests whether that discipline is useful.

Multimodal and diffusion post-training systems. VeRL-Omni extends veRL toward diffusion, unified, and omni-modality models and integrates vLLM-Omni rollout (Huang et al., 2026). Relax is explicitly omni-native (Zhang et al., 2026). Accordingly, “supporting image, video, or audio” cannot be the paper’s contribution. The comparison instead asks whether a single typed lineage simplifies cross-stage workflows and remains stable across execution modes.

Diffusion RL algorithms. DDPO casts denoising as a multi-step decision process (Black et al., 2023). Flow-GRPO converts flow sampling to an SDE and applies group-relative optimization (Liu et al., 2025); DanceGRPO generalizes related ideas across visual generation tasks (Xue et al., 2025); DiffusionNFT instead optimizes a forward-process objective from reward-ranked generations (Zheng et al., 2025). UniRL implements several algorithm families but does not claim their mathematical ideas. Its algorithm interface keeps their incompatible replay state explicit while reusing scheduling and training mechanics.

8 Limitations and Open Evidence

Integration is not automatic. Each new architecture still needs a bundle, stages, conditions, and adapters for the selected rollout runtime. A common trajectory type does not eliminate tokenizer, scheduler, latent-layout, LoRA-target, or serving-version incompatibilities.

Topology is configured, not optimized. Placement scopes make resource choices explicit but do not search for the best split. The best topology depends on model memory, reward cost, rollout variance, interconnect, and publication size.

Asynchronous execution is bounded and partially specialized. Publication is a barrier, and current AR and diffusion paths impose different admissible in-flight structure. Fault isolation is limited by the shared Ray runtime and role lifecycle; the system does not yet claim service-level recovery comparable to dedicated service architectures.

The empirical case is incomplete. At the audited commit, raw logs sufficient to verify existing benchmark prose were absent from the checkout. Therefore this draft makes no end-to-end throughput, scaling, memory, convergence, or final-quality claim. Sections 6.1–6.4 define the evidence required to remove this limitation.

9 Conclusion

UNIRL treats post-training as a trajectory program whose semantic object survives changes in generative family and execution topology. Lineage, typed modality values, replay segments, reward, credit, and policy version live in one IR; rollout, reward, training, transport, and publication are logical roles mapped separately to resources. This design is most valuable if it passes two empirical tests: credible learning behavior across representative AR and diffusion workloads, and competitive end-to-end execution without weakening semantic parity. Those tests are necessary but not sufficient for the paper’s trajectory-centric thesis: a real composed run must also demonstrate lineage-defined credit/grouping, and the measured IR/topology costs must bound the price of preserving that structure. The final paper will be complete only when artifact-backed evaluation establishes this full claim chain; bounded-staleness remains a conditional operating-region result rather than a prerequisite for every deployment.

Reproducibility statement

The source characterization in this draft refers to UniRL commit `f5d710406b215bb7a0b387fdd37e4d4778b92338` (2026-08-20). The manuscript repository records the claim-to-evidence plan in `PAPER_PLAN.md`. The cluster-ready execution protocol, configuration-parity gates, metric inventory, and immutable run-directory contract are specified in `EXPERIMENT_RUNBOOK.md`; the command-oriented transition to a fresh GPU-side agent is in `GPU_HANDOFF_README.md`. Both documents intentionally contain no results. The preliminary RQ3 evidence is regenerated by `experiments/trajectory_ir/run_cpu_evidence.py`; its manifest pins the source state, command, environment, workload, warm-up, repetitions, and timer. **TODO[data]: Before release, add the public artifact DOI/URL and author metadata, then archive manifests and regeneration scripts for every remaining result and figure.**

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A Claim-to-source audit

Table 8 records the executable source used to support mechanism descriptions. It does not prove performance or successful execution for every configuration.

Table 8: Primary code evidence at commit f5d7104.

Mechanism	Primary source symbols
Trajectory IR	<code>unirl/types/sample.py: Part, Sample; unirl/types/primitives.py; unirl/types/segments/</code>
Batch algebra	<code>unirl/distributed/tensor/batch.py: FieldKind, Batch</code>
Tensor references	<code>unirl/distributed/tensor/ref.py: TensorSpan, TensorRef; transport.py: TensorTransport</code>
Roles and placement	<code>unirl/distributed/group/device_pool.py, placement.py, handle.py, dispatch.py</code>
Rollout boundary	<code>unirl/rollout/engine/base.py: BaseRolloutEngine; concrete engines under unirl/rollout/engine/</code>
Training boundary	<code>unirl/algorithms/base.py: StageAlgorithm; unirl/train/stack/base.py: TrainStack</code>
Sync trainers	<code>unirl/trainer/ar.py: ARTrainer; unirl/trainer/diffusion.py: DiffusionTrainer</code>
Async scheduler	<code>unirl/trainer/async_rollout.py; unirl/rollout/manager/</code>
Weight publication	<code>unirl/distributed/weight_sync/; engine-specific receivers under unirl/rollout/engine/</code>
Instrumentation	<code>unirl/utils/profiling.py, memory_monitor.py, wandb_logger.py</code>

B Required run artifact schema

Every result row must resolve to a run directory containing:

- source and baseline commit hashes, submodule hashes, and dirty patches;
- the launch command and fully resolved configuration;
- model, reward, dataset, and tokenizer revisions or content hashes;
- GPU model/count, node/interconnect description, driver, CUDA, PyTorch, generation-engine, and training-backend versions;
- raw stdout/stderr, per-step structured timing, memory and utilization traces, checkpoint metadata, and evaluation outputs;
- seed, warmup exclusion rule, failure/retry accounting, and statistical aggregation script.

For training runs, serialized trace samples must redact private data while retaining lineage IDs, shapes, dtypes, policy versions, reward components, and segment index metadata. For system baselines, an alignment manifest must explain every nonidentical setting and whether it changes effective work.

C Correctness status and remaining checks

Executed locally. The preliminary RQ3 harness performs ten fail-fast checks on the audited source: lineage-valid cross-stage construction; exact split/concat; tree-complete select and slice; mean reward propagation plus group-normalized advantages; rejection of bad lineage, modality keys, and primitive batch sizes; packed variable-length text select/concat; and lazy tensor-reference selection, materialization, identity, and bounds behavior. The structured outcomes and data-sensitive SHA-256 round-trip hash are stored with the run artifacts.

Required before performance interpretation. The following system-level checks remain prerequisites for RQ1, RQ2, and RQ4:

1. **Rollout-replay parity:** before the first update, compare engine-emitted and train-side replay log probabilities or transition statistics under stated numerical tolerances.
2. **Publication parity:** repeat the parity check after each LoRA or full-weight publication path used in an experiment.
3. **Batch-version integrity:** verify that an optimizer batch never mixes behavior versions when the configured algorithm assumes one version.

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4. **Boundary integrity:** checkpoint and evaluation begin only after the async manager is empty; resume restores the optimizer version and republishes it before new rollout.
 5. **Work accounting:** compare generated tokens or pixels, denoising and replay steps, scored samples, and committed optimizer updates across systems.